Multi-Agent Architecture

https://github.com/aiyshwary/multi-agent-revenue-churn

Python LLM Multi-Agent Pandas Docker GitHub Actions

OVERVIEW

A production-grade agentic pipeline that combines LLM-driven planning with deterministic Python analytics to perform revenue analysis and churn prediction, featuring an Orchestrator-Executor-Reflection loop with memory management and Docker deployment.

IMPACT

Designed and implemented a multi-agent revenue and churn analysis system where an LLM Planner coordinates an Executor, Validator, and Reflection agent, keeping LLMs focused on reasoning while Python handles all numeric computation.

KEY DETAILS

- Planner emits an executable plan (list or graph of nodes) from a high-level task description.
- Orchestrator runs each step via the Executor, applies deterministic validators, and uses Reflection to decide retry or replan actions.
- Memory Manager persists short summaries and semantic snippets for context retrieval across long pipelines.
- Outputs churn_report.json, per-client quarterly revenue CSVs, and HTML/PNG visualizations per run.
- Fully containerized with Docker Compose; CI/CD via GitHub Actions with pytest unit test suite.